



Bangladesh University of Business and Technology

Course Title: Computer Networks Lab

Course Code: CSE 320

LAB REPORT

Experiment Name: HTTP Web Server Accessing with DNS and FTP Server

Experiment No.: 02

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Introduction

HTTP, DNS, and FTP are distinct application-layer protocols that work together to enable various network functions.

DNS (Domain Name System): DNS translates human-readable domain names (e.g., *www.example.com* or *ftp.example.com*) into the numerical IP addresses that computers use to communicate. This is the foundational step for both HTTP and FTP access.

HTTP (Hypertext Transfer Protocol): HTTP is used for accessing and viewing websites in a web browser. The browser uses the IP address obtained from DNS to send HTTP requests to a web server, which then sends back web pages and content.

FTP (File Transfer Protocol): FTP is used specifically for transferring files between a client and a server (uploading and downloading), which is distinct from the typical web browsing function of HTTP. It is commonly used for website administrators to upload their website files to a web server.

Devices and Tools

- Cisco Packet Tracer v9.0.0
- 2911 Router
- 2960-24TT Switch × 3
- Client PC × 12
- Server (DNS, HTTP, FTP)

Addressing Table

Device	IP Address	Subnet Mask	URL
PC0	192.168.10.2	255.255.255.0	-
PC1	192.168.10.3	255.255.255.0	-
PC2	192.168.10.4	255.255.255.0	-
PC3	192.168.10.5	255.255.255.0	-
PC4	192.168.20.2	255.255.255.0	-
PC5	192.168.20.3	255.255.255.0	-
PC6	192.168.20.4	255.255.255.0	-
PC7	192.168.20.5	255.255.255.0	-
PC8	192.168.30.2	255.255.255.0	-
PC9	192.168.30.3	255.255.255.0	-
PC10	192.168.30.4	255.255.255.0	-
PC11	192.168.30.5	255.255.255.0	-
DNS Server	192.168.30.6	255.255.255.0	www.bubt.com
HTTP Server	192.168.10.6	255.255.255.0	www.bubt.com
FTP Server	192.168.20.6	255.255.255.0	ftp.bubt.com

Procedure

The following steps were meticulously followed to setup an HTTP web server with FTP:

Preparing the Network:

- In the Cisco Packet Tracer application, we add a new **Router (2911)** and three **Switches (2960-24TT)**. Each of the Switches are connected to the Router.
- Now each **Switch** is connected to four **client PCs** and one **Server**. There are three **Servers**: one for DNS, one for HTTP, and one for FTP.
- The network devices are connected using the **Auto Cabling** tool in the Cisco Packet Tracer application.

Configuring the Router:

- Single click the **Router0** and a new window appears.
- Under the **Interfaces** tab, we observe three interfaces: **GigabitEthernet0/0**, **GigabitEthernet0/1**, **GigabitEthernet0/2**. We check the **Port Status** checkbox to turn on all three interfaces.
- Also, under the **Interfaces** tab, we configure the **IPv4 Addresses** for the available interfaces (the **Subnet Masks** will be assigned automatically):

Interface	IPv4 Address
GigabitEthernet0/0	192.168.10.1
GigabitEthernet0/1	192.168.20.1
GigabitEthernet0/2	192.168.30.1

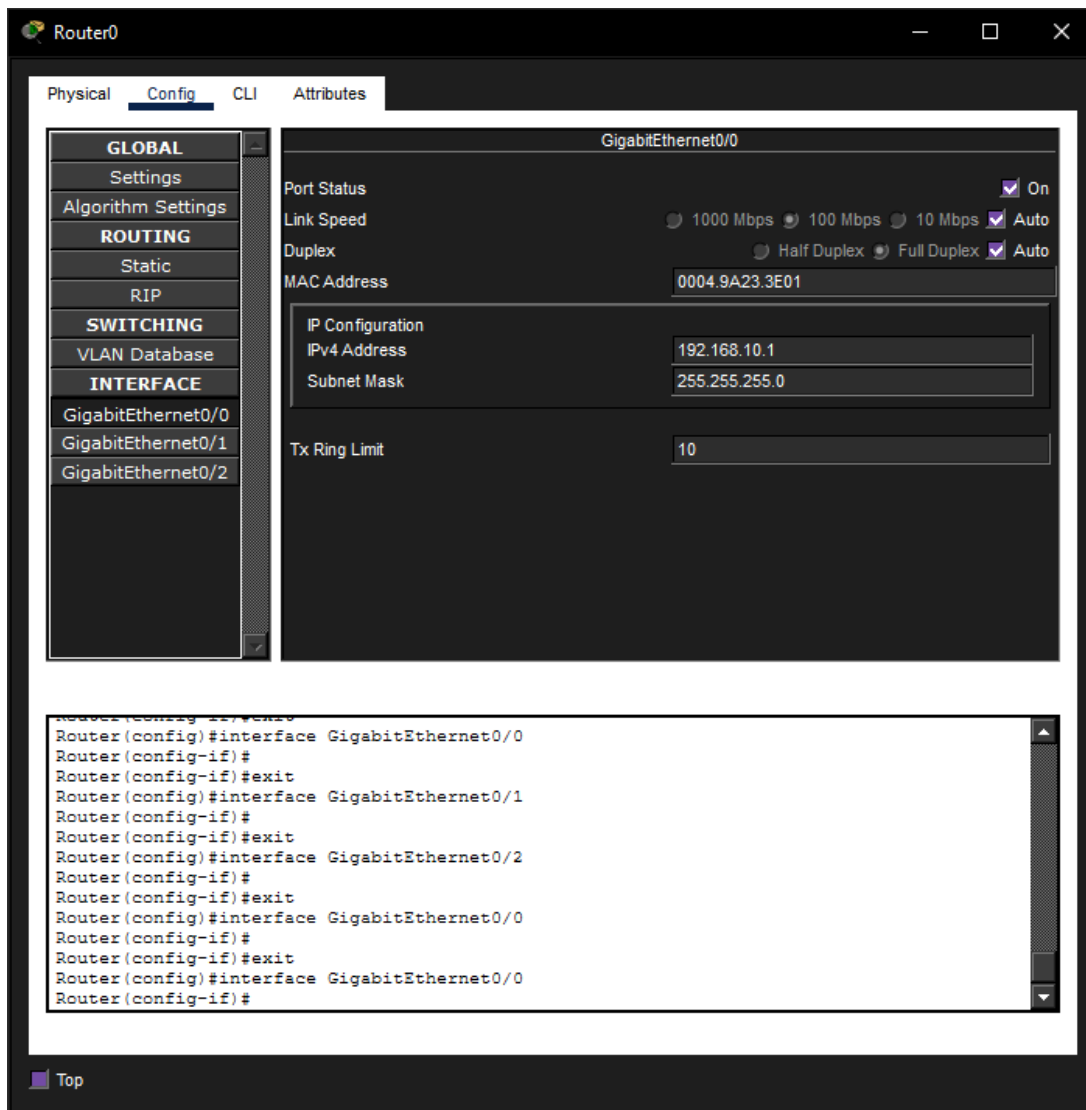


Fig 1.1: Configuring the Router.

Configuring the Clients and Servers:

- Set up the **IPv4 Addresses** of PC0-PC11, DNS, HTTP, and FTP servers according to the aforementioned addressing table.
- Set up the **Default Gateway** address and **DNS server** according to the following table:

Devices	Default Gateway	DNS Server
PC0-PC3, HTTP Server	192.168.10.1	192.168.30.6
PC4-PC7, FTP Server	192.168.20.1	192.168.30.6
PC8-PC11, DNS Server	192.168.30.1	192.168.30.6

- **Setting up individual devices:**
 - Click the **Client** or **Server** device. A new window appears.
 - Under the **Desktop** tab, click **IP Configuration** and set the IPv4 Address, Default Gateway, and DNS Server.

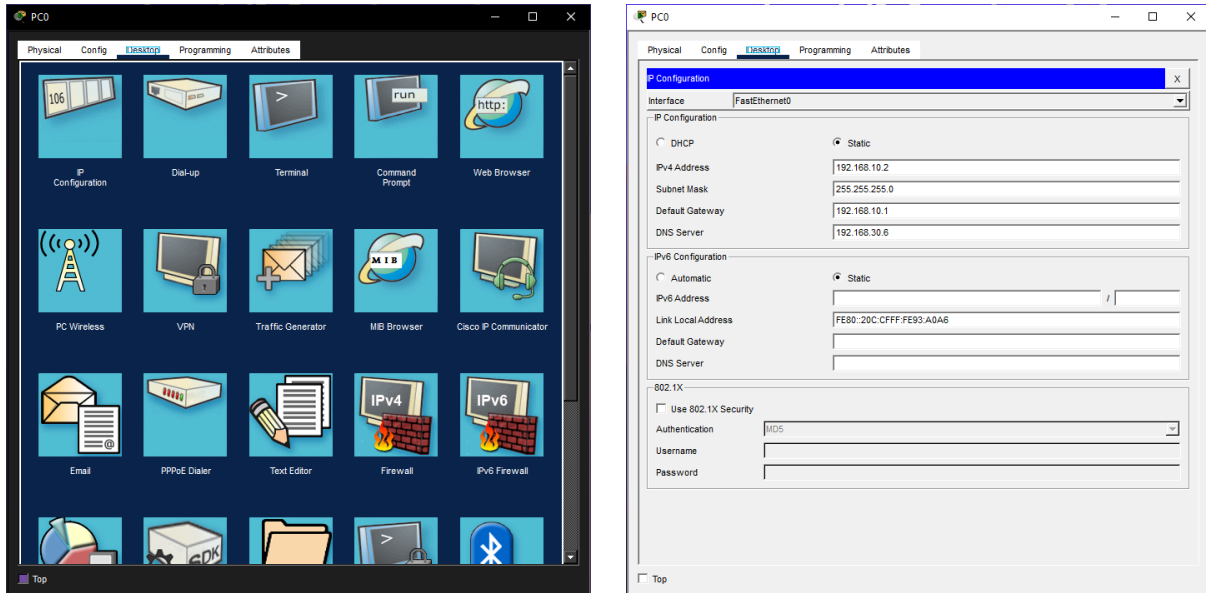


Fig 1.2: Configuring client and server devices.

Configuring the DNS Server:

- Click the **DNS server**, a new window will appear.
- Go to the **Services** tab and select **DNS**. Turn on the **DNS Service**.
- Under the **Resource Records**, type **bubt.com** in the Name field.
- In the Address field, type the IPv4 address of the **HTTP Server** which is mentioned in the addressing table.
- Click **Add** to add a new **Address Record**. This ensures that our DNS server points to the HTTP server to serve the desired web page.

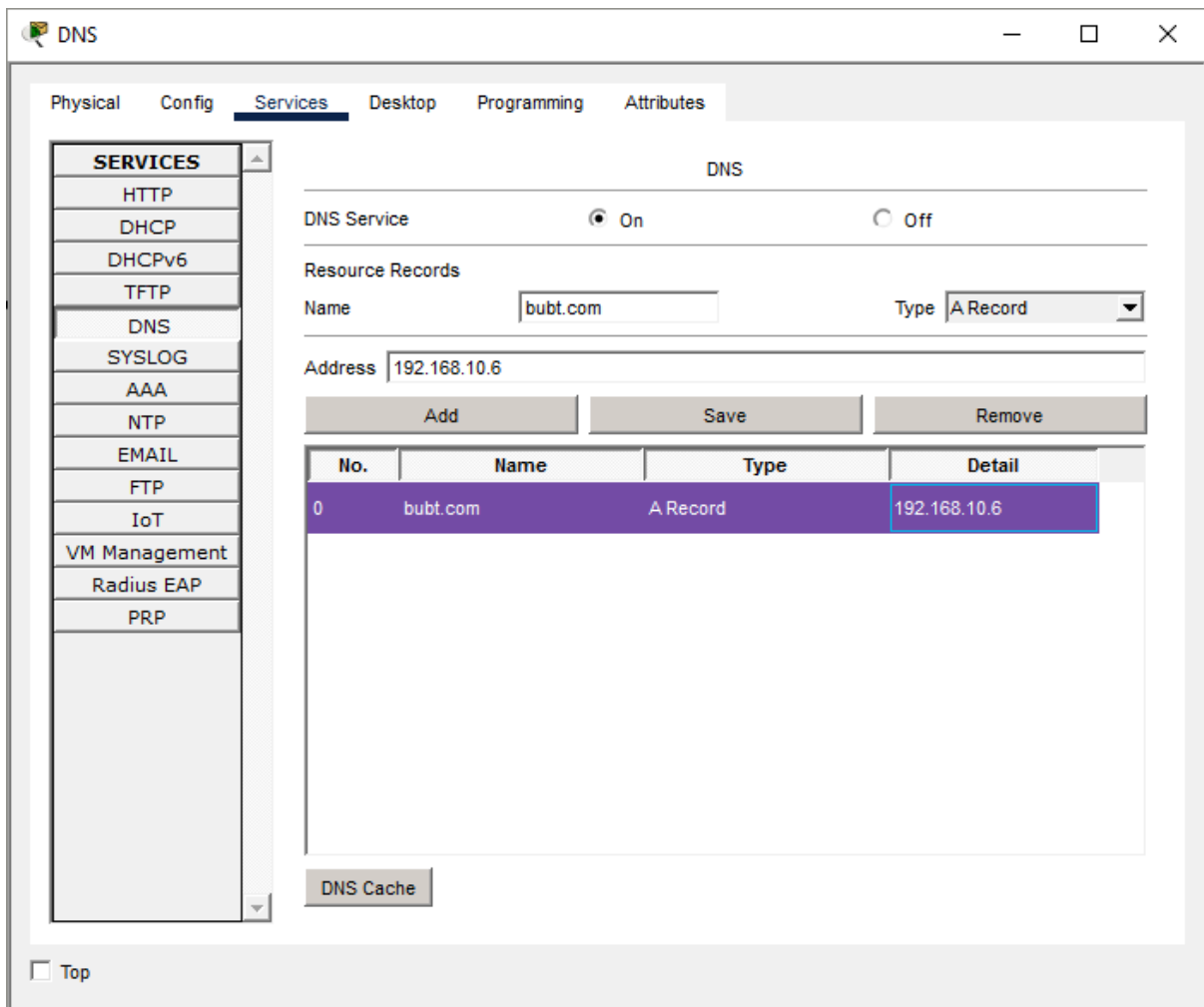


Fig 1.3: Configuring the DNS Server.

Configuring the HTTP Server:

- Click the **HTTP server**, a new window will appear.
- Go to the **Services** tab and select **HTTP**. Turn on the **HTTP and HTTPS Service**.
- Under the File Manager, select **index.html** and click **(edit)**. Enter the following **HTML code** and save the file:

```
<html>
    <h1>This is Working</h1>
</html>
```

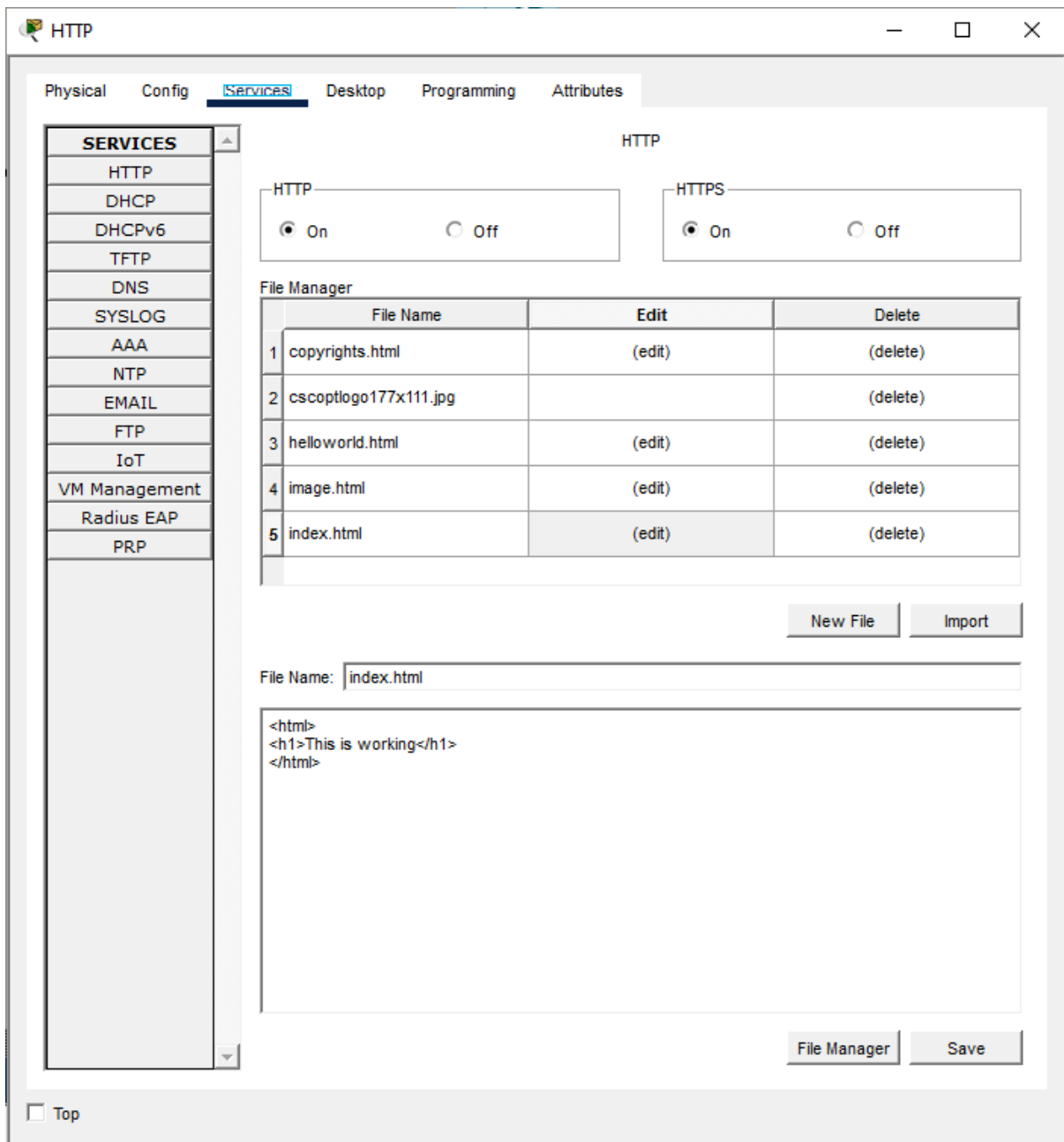


Fig 14: Configuring the HTTP Server.

Configuring the FTP Server:

- Click the **FTP server**, a new window will appear.
- Go to the **Services** tab and select **FTP**. Turn on the **FTP Service**.
- Under the **User Setup**, type **bubt** in both Name and Address fields.
- Check **Write**, **Read**, **Delete**, **Rename**, and **List** permissions and click **Add**. This ensures that we would be able to login to the FTP server using the credentials we just added.

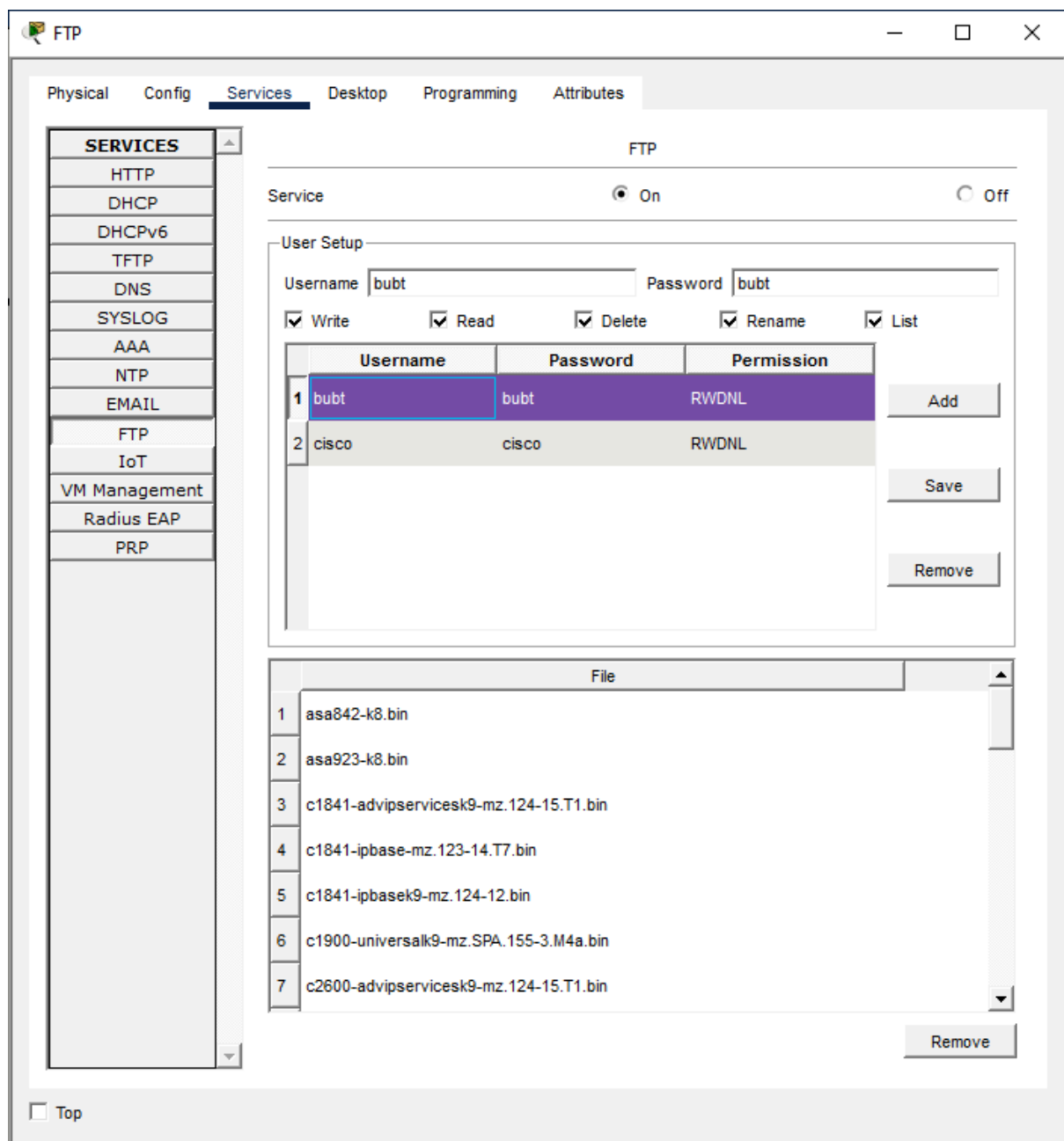


Fig 1.5: Configuring the FTP Server

Uploading a file to the FTP Server:

- Click the **FTP server**, a new window will appear.
- Go to the **Desktop** tab and select **Text Editor**. Type “BUBT” in the text field. Hit **CTRL + S** to save the file.
- A new prompt will appear to enter the new file’s name. Type **ONE_file.txt** and click **OK**.
- Close the **Text Editor** and select **Command Prompt**.

- Login to the FTP server using the credentials.

```
C:\>ftp bubt.com
```

```
Trying to connect... bubt.com
```

```
Connected to bubt.com
```

```
220- Welcome to PT Ftp server
```

```
Username: bubt
```

```
331- Username ok, need password
```

```
Password: ****
```

```
230- Logged in
```

```
(passive mod On)
```

```
ftp>
```

- Type the following command to upload the newly created file to the FTP Server:

```
ftp>put ONE_FILE.txt
```

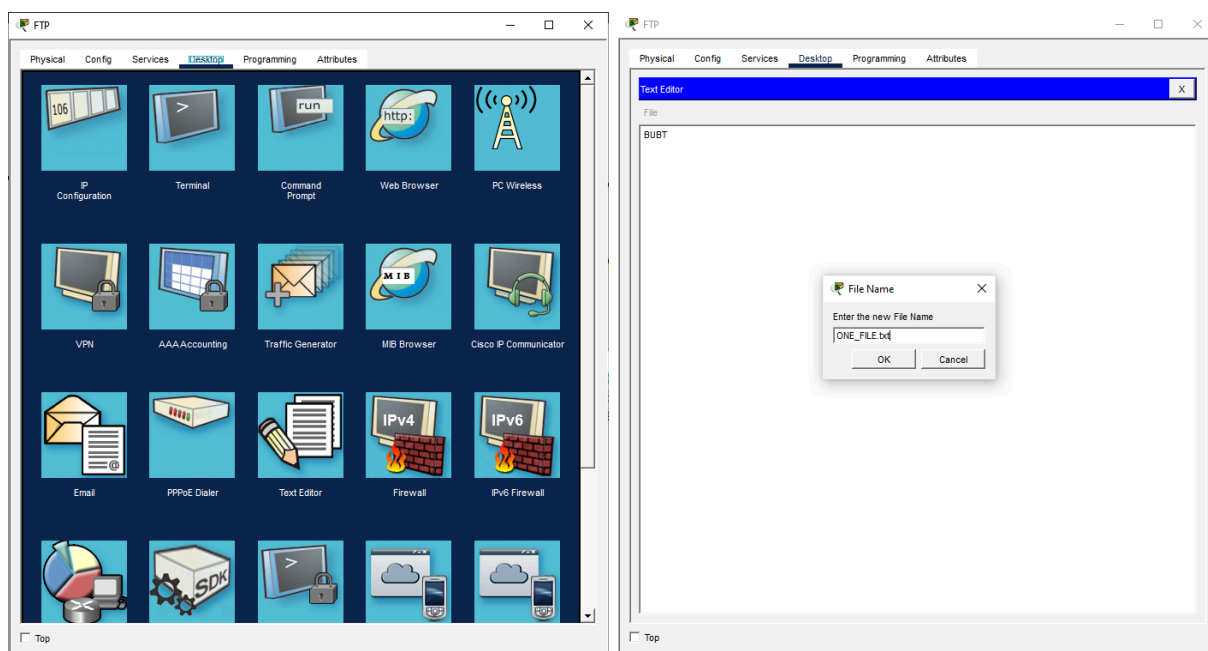


Fig 1.6: Creating a new text file.

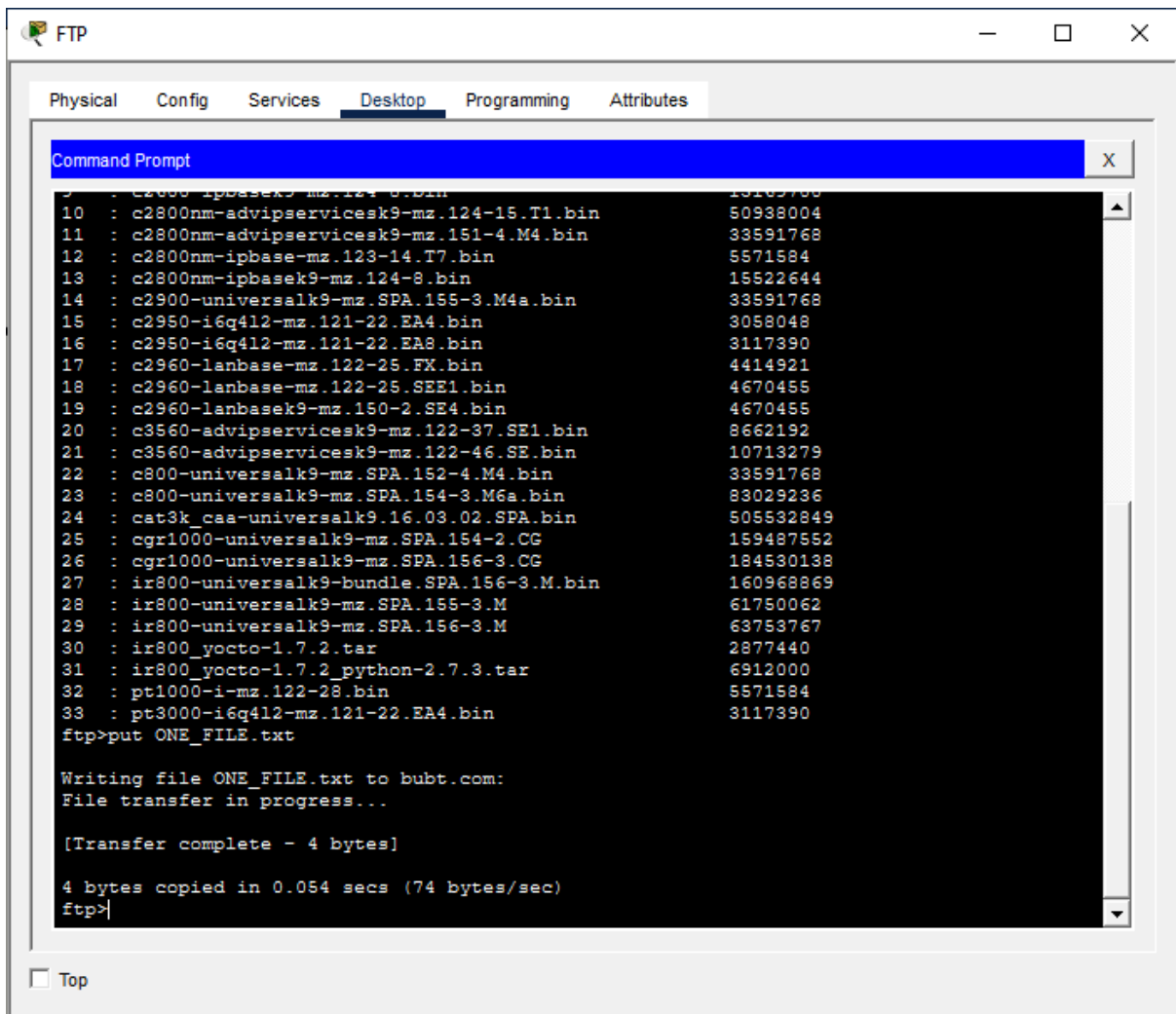


Fig 17: Uploading the text file to the FTP Server.

Result

Once we have followed all the steps mentioned in the previous section, we should have a working network consisting of 1 Router, 3 Switches, 12 Clients, and 3 Servers (DNS-HTTP-FTP):

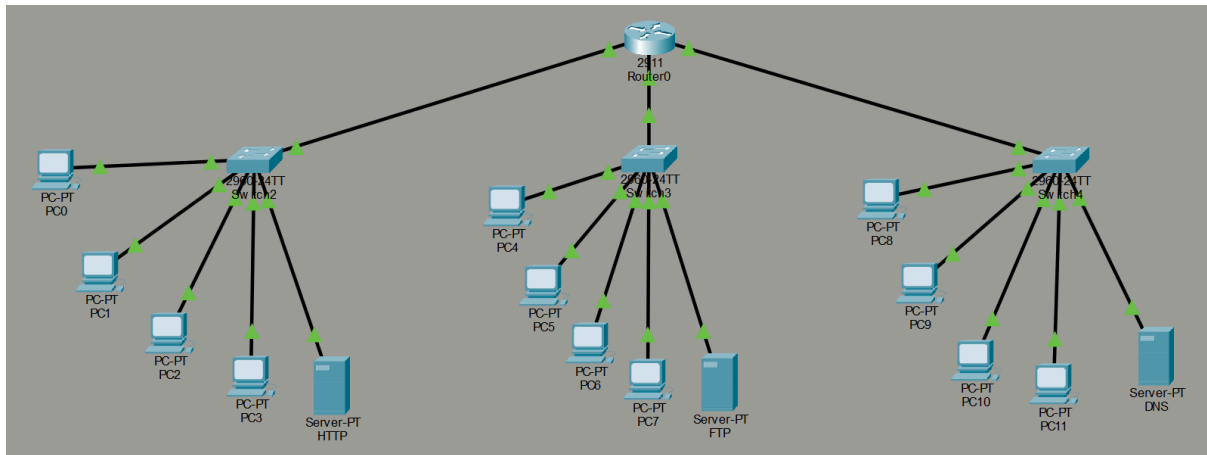


Fig 2.1: HTTP Web Server with DNS and FTP.

Testing and Verification

Since we are using the same DNS server, we only need to test the HTTP and FTP servers to ensure all three servers are working perfectly.

Testing the HTTP Server:

- Click on any of the client PCs, a new window will appear.
- From the **Desktop** tab, click on **Web Browser** (it might take a while to load)
- In the URL field, type **bubt.com** and click Go.
- If we see the text appearing “**This is working**”, the HTTP and DNS servers are OK.



Fig 3.1: Testing the HTTP Server.

Testing the FTP Server:

- Click on any of the client PCs, a new window will appear.
- From the **Desktop** tab, click on **Command Prompt**
- Login to the FTP server located at **bubt.com**
- Use **get ONE_FILE.txt** command to download the file from the FTP Server to the Client.

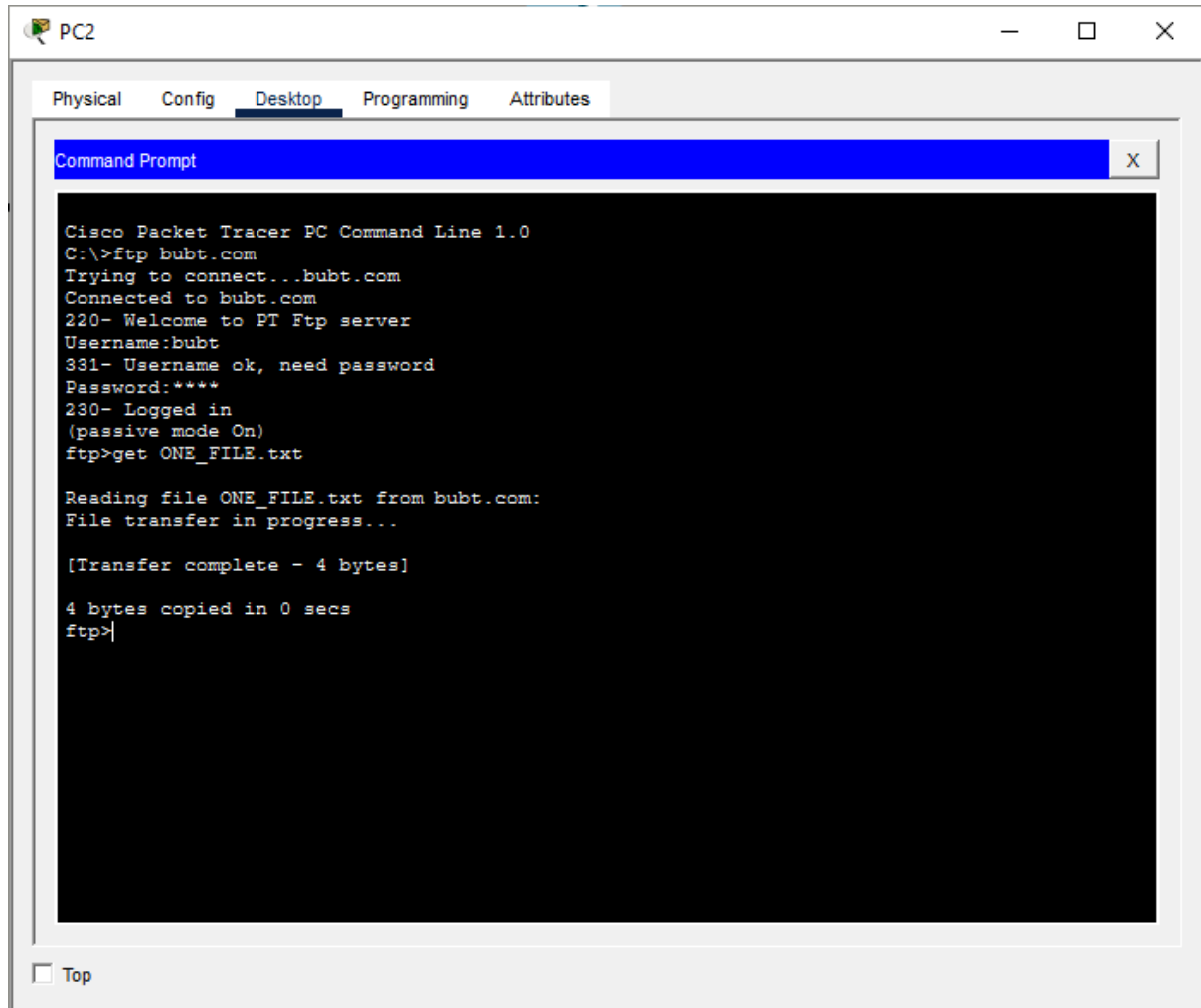


Fig 3.2: Testing the FTP Server.

Conclusion

The objective of the second experiment was successfully achieved by building an HTTP web server with DNS and FTP server. It provided core skills for client-server network management.